Ghislain THEUBO

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LinkedIn | Bilingual English/French | GitHub

Professional Summary

Software Engineer with solid **C++**, **algorithms**, and **software architecture** skills, plus firsthand R&D experience at **Dassault Systèmes** on the **3DEXPERIENCE** platform. Strong at rigorous development, performance tuning, and turning specifications into robust converters/APIs. Built personal projects on **data format conversion**, **schema mapping**, and high-throughput pipelines — aligned with interoperability bridges for Aerospace & Defense and Transportation customers. Basic **JavaScript**; interest in **AI-assisted** data transformation. Eager to join ENOVIA Interoperability R&D in Aix-en-Provence.

Core Strengths

C++ (OOP, STL, templates) · Algorithms & architecture · Data conversion / interoperability · API quality & performance · Specifications & documentation · JavaScript (basics) · Debugging & profiling · English (written & oral) · Autonomy & rigor

Experience

Dassault Systèmes – 3DEXPERIENCE R&D Vélizy-Villacoublay, France
Software Engineer (Platform Architecture & Performance) *Apr 2025 – Oct 2025*

- Contributed to platform features in a 20-person R&D team; participated in **technical specifications**, design discussions, and cross-team delivery.
- Improved critical service performance by **35%** through algorithmic optimization and careful API/data-path design — mindset transferable to **high-volume data conversion**.
- Maintained quality via **unit/integration tests** (85% coverage), CI gates, and structured technical writing for maintainability.
- Collaborated in an **international** product environment; practiced clear English communication for design and reviews.

INRIA Grenoble – Humanlab / Froll'N'Roll Montbonnot-Saint-Martin, France
Research Intern (Real-Time Systems & C/C++) *May 2024 – Aug 2024*

- Developed **C/FreeRTOS** and Python modules processing sensor streams under latency constraints; redesigned protocols for **+40%** responsiveness.
- Applied OOP and efficient data structures; validated designs with end users and contributed to documented, open-source code.

Gustave Roussy Cancer Institute & CentraleSupélec MICS Lab Paris, France
Backend Engineer (CDD – APIs & Data Pipelines) *Jan 2026 – Present*

- Built production APIs with validation, access control, and observability; optimized SQL/indexing for faster data exchange (**30%**).
- Automated packaging and delivery (Docker, GitHub Actions); wrote operational documentation for data flows.

Laboratoire d'Informatique de Grenoble (LIG) Grenoble, France
Cloud Infrastructure Intern *Feb 2025 – Mar 2025*

- Studied migration/performance trade-offs on OpenStack/Grid'5000 — analytical approach useful for compatibility and volumetry topics.

Personal Projects – Interoperability, Conversion & C++

Multi-Format Engineering Data Converter | C++, CMake, Catch2, JSON/XML GitHub

- Built a **C++ conversion toolkit** mapping between simplified engineering schemas (JSON ↔ XML ↔ proprietary CSV) with validation and error reporting — mirrors interoperability "bridges" for customer data exchange.
- Focused on **robustness** (malformed input), **compatibility** (versioned schemas), and **performance** on large files (streaming parsers, memory profiling).
- Wrote clear specifications + architecture notes; exposed a thin **JavaScript** demo UI calling the converter via a local CLI/API wrapper.

High-Volume Batch Transform Pipeline | C++, multithreading, benchmarks GitHub

- Designed a pipeline for batch transformation of large datasets with worker pools, back-pressure, and progress metrics — addressing **volumetry** and **performance** challenges.
- Compared sequential vs parallel strategies; documented bottlenecks and proposed architectural evolutions.

C++ Polyhedral Mesh Library (KD-Tree & Graphs) | C++, Templates, Catch2 GitHub

- Header-only mesh library with KD-Tree queries and graph algorithms — demonstrates deep C++ craftsmanship and algorithmic rigor.

Technical Skills

Languages: **C++** (OOP, STL, templates), C, Python, Java, **JavaScript (basics)**, SQL, Bash
Interoperability: schema mapping, format conversion, validation, versioning, API bridges
Architecture: modular design, performance analysis, concurrency, documentation of technical choices
Quality: Catch2, unit tests, CI, debugging, profiling, code reviews
Interest: AI-assisted transformation / validation of engineering data
Environment: Linux, Git, CMake, Docker; prior **Dassault Systèmes** / **3DEXPERIENCE** R&D experience

Education

Grenoble INP - Ensimag — Grenoble, France
Engineering Degree in Computer Science & Software Engineering — *2023 – 2025*

École Nationale Supérieure Polytechnique de Yaoundé — Yaoundé, Cameroon
Master's Degree in Computer Engineering (Top 12%) — *2019 – 2023*

Languages

English: Bilingual — strong written & oral communication (international collaboration)
French: Fluent
Spanish: Intermediate